

MATH 3551 A — Abstract Algebra I

Practice Midterm 2

Taylor Dupuy, University of Vermont • Fall 2026

For practice — Midterm 2 is Friday, November 6, 2026 • Time limit: 50 minutes • Total: 100 points

NAME (please print): _____

Directions. This exam is **closed book**: no textbook, no notes, no calculators, no phones, no other devices, and no assistance of any kind. Write your answers on the exam itself; use the backs of pages if you need more room, and say clearly when you have done so. *Show your work and name the theorems you use* — “by Lagrange’s Theorem, $|H|$ divides 24” earns nearly all of the credit, an unjustified correct number earns little. Point values are marked on each problem. Good luck.

Problem 1. (30 points) **True or false; prove or disprove.**

Circle **T** or **F**, and then give a one-line reason: a theorem, a short computation, or an explicit counterexample. Three points each.

- (a) **T** **F** If $H \leq G$ and $N \leq G$, then HN is a subgroup of G .
- (b) **T** **F** If $K \trianglelefteq H$ and $H \trianglelefteq G$, then $K \trianglelefteq G$.
- (c) **T** **F** The symmetric group S_5 has exactly seven conjugacy classes.
- (d) **T** **F** The automorphism group $\text{Aut}(\mathbb{Z}/8\mathbb{Z})$ is cyclic.
- (e) **T** **F** If $|G| = p^3$ for a prime p , then $Z(G) \neq 1$.
- (f) **T** **F** The alternating group A_4 has a normal subgroup of order 4.
- (g) **T** **F** For $n \geq 3$, every permutation in S_n is a product of 3-cycles.
- (h) **T** **F** If a finite group G acts transitively on a set A , then $|A|$ divides $|G|$.
- (i) **T** **F** Every group of order p^2 , p prime, is cyclic.
- (j) **T** **F** $\text{Inn}(G)$ is a normal subgroup of $\text{Aut}(G)$.

Problem 2. (20 points) **An action of A_4 .**

Let $G = A_4$ act on the set $A = \{P_1, P_2, P_3\}$ of the three ways of splitting $\{1, 2, 3, 4\}$ into two unordered pairs:

$$P_1 = \{\{1, 2\}, \{3, 4\}\}, \quad P_2 = \{\{1, 3\}, \{2, 4\}\}, \quad P_3 = \{\{1, 4\}, \{2, 3\}\},$$

where σ acts on a pair by $\sigma \cdot \{i, j\} = \{\sigma(i), \sigma(j)\}$ and on a splitting by acting on each of its two pairs.

- (a) (6 points) Determine the permutations of A induced by $(1\ 2\ 3)$ and by $(1\ 2)(3\ 4)$.
- (b) (6 points) Let $\varphi: A_4 \rightarrow S_3$ be the resulting permutation representation. Determine $\ker \varphi$ and $\text{im } \varphi$, with proof.
- (c) (4 points) Deduce that $V_4 = \{1, (1\ 2)(3\ 4), (1\ 3)(2\ 4), (1\ 4)(2\ 3)\}$ is a normal subgroup of A_4 and identify the quotient A_4/V_4 .
- (d) (4 points) Determine the stabilizer of P_1 in A_4 and verify the orbit–stabilizer relation.

Problem 3. (15 points) **Conjugacy in S_7 .**

- (a) (6 points) Determine the number of elements of order 4 in S_7 .
- (b) (5 points) Determine the size of the conjugacy class of $(1\ 2\ 3\ 4)(5\ 6)$ in S_7 .
- (c) (4 points) Determine the order of $C_{S_7}((1\ 2\ 3\ 4)(5\ 6))$ and describe its elements explicitly.

Problem 4. (20 points) **Actions on cosets.**

Let G be a finite group and let $H \leq G$ with $|G : H| = n$. Let G act on the set of n left cosets of H in G by left multiplication, $g \cdot (xH) = gxH$, and let $\pi : G \rightarrow S_n$ be the resulting permutation representation.

- (a) (6 points) Prove that $\ker \pi \leq H$.
- (b) (8 points) Prove that $\ker \pi$ contains every normal subgroup of G that is contained in H , and that $|G : \ker \pi|$ divides $n!$.
- (c) (6 points) Suppose now that $|G| = 48$ and that G has a subgroup H with $|G : H| = 4$. Prove that G is not simple.

Problem 5. (15 points) **Composition series.**

- (a) (3 points) Define *composition series* and *composition factors*.
- (b) (5 points) Give a composition series for $D_{12} = \langle r, s \mid r^6 = s^2 = 1, rs = sr^{-1} \rangle$ and list its composition factors, in order.
- (c) (5 points) Give a second composition series for D_{12} , different from the one in (b), in which the composition factors appear in a *different order*. What does the Jordan–Hölder Theorem predict about the two lists?
- (d) (2 points) Is D_{12} solvable? Justify your answer from (b).

End of practice exam. Total: $30 + 20 + 15 + 20 + 15 = 100$ points.